

Optimization of the Overlap Rate of Multi-track Laser Cladding Based on the Flat-Top Overlapping Model

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Abstract The overlap rate of multi-track laser cladding determines the surface flatness of the cladding layer. To improve the surface flatness of the cladding layer, an optimization method for the overlap rate of multi-track laser cladding layer based on the flat-top overlapping model is proposed. Firstly, the cross-section profile data of single-track laser cladding are collected, and curve fitting is performed to obtain the optimal fitting cross-section fitting function. Then, the multi-track overlapping model is established based on the optimal fitting function, and the optimal overlap rate model for the multi-track laser cladding is derived theoretically. Finally, the optimization method of the overlap rate is verified experimentally, using 20Cr13 as the substrate, M2, 316L and 15-5PH as the coating powder respectively. The experimental results showed that the flatness of the multi-track cladding layers with different cladding powders is greater than 85% with minor errors. It is shown that the model is helpful to obtain good surface flatness and is suitable for various powders, and the model has good effectiveness and versatility.

Keywords Additive manufacturing · Laser cladding · Overlapping model · Curve fitting

1 Introduction

Additive manufacturing (AM) is a competitive technology with advantages such as high material utilization and short manufacturing time compared to traditional subtractive manufacturing such as turning, milling and grinding. Laser cladding is a new additive manufacturing technology using a high-energy laser beam as the heat source, with the advantages of a small heat-affected zone [1, 2], low dilution rate [3], and good metallurgical bonding between the cladding layer and the substrate [4]. It provides a solution for preparing high performance coatings and repairing complex parts, which has received more and more attention from academia and industry in recent decades, such as aerospace and mining machinery [1, 5].

Despite the many advantages of laser cladding technology, the surface flatness of the cladding layer still needs considerable improvement. The surface flatness of the multi-track laser cladding process directly affects the processing allowance and the difficulty of subsequent processing, and the surface flatness of the multi-track laser cladding process is mainly influenced by the overlap rate [6, 7]. However, the current research has mainly focused on laser power, scanning speed and powder feeding speed while ignoring the overlap rate. When the overlap rate is too low, it will increase the surface roughness of the multi-track cladding layer and will produce defects such as holes and cracks. Conversely, the excessive overlap rates will damage the formed surface of the overlap area and reduce the flatness of the overlap surface. Therefore, a suitable overlap rate needs to be determined to improve the surface flatness of the cladding layer.

To determine the appropriate overlap rate, a reasonable overlapping model is required. A proper overlap rate ensures a good surface flatness for the forming cladding.

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Among the mathematical models describing the morphology of multi-track cladding layers, the geometric model is a way to identify the optimal overlap rate [7]. Therefore, many studies have been conducted on overlapping models in additive manufacturing. Cao et al. [8] used Gaussian, logistic, parabolic and sine functions to curve fit the weld bead cross-section profile, and selected the sine function to establish the overlapping model applicable to gas metal arc welding (GMAW) and performed the calculation of the optimum overlap rate. Xiong et al. [9] used the parabolic function, circular arc function and cosine function to curve fit the cross-section profile of the weld bead, based on which the circular arc function was selected to establish a flat-top overlapping model applicable to the GMAW overlapping model.

In addition to using the flat-top overlapping model, the tangent overlapping model is another commonly used one. Ding et al. [10] proposed a tangent overlapping model to calculate the optimal center distance of adjacent weld beads based on the ideal overlap condition with a single-track profile function as a parabolic function. After that, the overlapping model was combined with an artificial neural network (ANN) to achieve adaptive path planning for multi-track overlap [11]. Fang et al. [12] used the parabolic function, cosine function and circular arc function for contour fitting of single weld bead and selected the parabolic function to establish a single layer multi-track slant-top overlapping model applicable to cold metal transition arc welding; experimentally verified the validity of the overlap parameters of this model to improve the morphology of multi-track formed parts.

Beyond the conventional overlapping model, many scholars have adopted different overlapping strategies to improve the surface morphology of the formed layers. Li et al. [13] used an artificial neural network (ANN) to predict the offset distance of the weld bead and developed an overlapping model considering the weld bead melting extension. The validity of the model was experimentally verified, and the surface flatness of multi-track GMAW was improved. After that, they proposed a layer overlapping strategy for cuboid [14] and inclination [15] multi-layer multi-track based on this. Ocelík et al. [16] proposed a recursive model for laser cladding to predict the profile of the cladding layer overlap forming section, which predicts the cladding height and the ripples of the cladding layer. It was experimentally verified to agree with the experimental results within a specific range of process parameters. Chen et al. [17] proposed a recursive model based on coordinate transformation and verified the feasibility of the model through experiments. The experimental results show that the model is effective in predicting morphology and can effectively improve molding quality. Hu et al. [18] developed a variable section overlapping model applicable to GMAW, fitting a second weld bead profile

with a rotationally varying parabolic function and a circular arc function, predicting the overlap weld bead profile and calculating the optimal overlap center distance. From the aforementioned literature, it can be concluded that, on one hand, the study of overlapping models is mainly applied in fields such as GMAW, while there are fewer studies on overlapping models for laser cladding. On the other hand, most of the models in the literature related to overlapping models are only applicable to a single material. Therefore, it is necessary to propose an overlapping model suitable for laser cladding and various cladding powder.

In this study, based on the flat-top model, an overlapping model applicable to the laser cladding of different cladding powders is established, which lay the foundation for the subsequent research work on the preparation of high-quality multi-layer multi-track cladding layers. The rest of the paper is structured as follows. In Sect. 2, full factorial experiments were performed, and cladding layer profile data were obtained, followed by curve fitting of the cladding layer profile. In Sect. 3, the overlap rate model applicable to laser cladding is developed by combining the fitting results of Sect. 2 and the flat-top model. In Sect. 4, verification experiments with various cladding powders are performed, and the experimental results are given. Finally, the conclusions are shown in Sect. 5.

2 Cross-Section Profile Model

2.1 Single-Track Empirical Models

The geometric modeling of the single-track cladding layer cross-section profile is shown in Fig. 1, where h represents the cladding height of the single-track cladding layer, and w represents the cladding width of the single-track cladding layer. To establish a concise mathematical model for the cross-section profile of the single-track cladding layer, parabolic, cosine and Gaussian functions were selected to fit the single-track cross-section profile by combining the cross-section profile geometric model and some scholars'

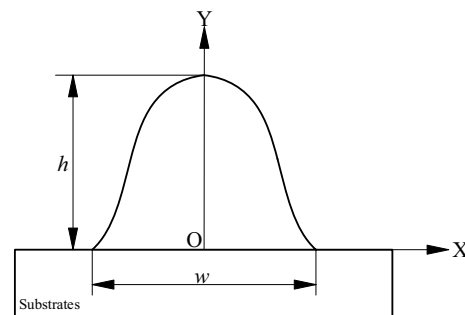


Fig. 1 Geometric model of cross-section profile

previous studies [8, 9, 11, 18], and the expressions of the selected functions are shown in Table 1.

2.2 Experimental Design and Equipment

The purpose of this experiment is to measure the cross-section profile data of the single-track cladding layer. The effects of laser power, scanning speed and powder feeding speed amount are not independent. To obtain sufficient data to improve the accuracy of the model, a three-factor and five-level full factorial experimental design (DOE) was used. 20Cr13 was used as the substrate and 15-5PH powder was used as the cladding powder for this full factorial experiment. The range of values of each factor directly affects the cross-sectional profile of the cladding layer [19, 20], and the experimental process parameters are selected based on the previous experimental research of the research group [21]. The full factorial experimental design is shown in Table 2.

The laser cladding equipment for this experiment is shown in Fig. 2a, which consists of the laser head, powder feeder, six-axis robot, and water cooler. A Leica DVM6S 3D super depth-of-field microscope (as shown in Fig. 2b) was used for data acquisition of the specimens.

2.3 The Results of Curve Fitting

The surface morphology of the single-track cladding layer is shown in Fig. 3, and it can be seen that there is a clear curved texture on the cladding layer, and the texture has a clear directionality. This is due to the metal flow resulting from the melting of the metal during the cladding process. Therefore, the oblique line value measurement is used to measure the contour of the single-track cladding layer to make the value range include all texture areas. The measurement schematic is shown in Fig. 3.

To ensure the uniformity and continuity of the measurement area of the cladding layer, the middle section of the

cladding layer was selected to measure by the ultra-depth-of-field microscope, and the coordinate data of the cross-section profile are derived with the center of the cladding layer as the coordinate Origin.

The measured cross-section profile data is fitted with the selected function by Origin. The comparison between the fitted curve and the measurement data is shown in Figs. 4, 5 and 6.

The adjusted R^2 is an evaluation index reflecting the degree of fit of the function, and the closer it is to 1, the better the fit is. It does not get closer to 1 with the independent variable increase. The residual sum of squares SSE is the sum of squares of the difference between the fitted curve and the measurement data. The closer the SSE is to zero, the better the fit is. The significant F value is also a commonly used evaluation index of curve fitting. The larger the F value, the more significant the effect of the fitting. Some of the fitting results are shown in Table 3.

In the curve fitting, the adjusted R^2 is generally used as the main evaluation index to select the optimal model among them. As seen from Table 3, the adjusted R^2 of the Gaussian function is better than the other two functions. The significance F -value and adjusted R^2 have been found to vary in the same direction. The significance F -value of the fitting result of the Gaussian function is significantly larger than the fitting result of the parabolic function and cosine function. To ensure higher accuracy of the model, the Gaussian function with better fitting results was selected for the subsequent single-layer multi-track model building. And the 106th group of parameters ($P = 1500\text{W}$, $V = 11\text{ mm/s}$, $F = 1.9\text{ r/min}$) with the largest adjusted R^2 was selected as the parameters for the subsequent validation experiments. The results of the function fitting for the 106th group of parameters are shown in Table 4.

3 Methodology

3.1 Results of Different Overlap Rates

To obtain a large area of the laser cladding layer, this needs to be achieved by mutual lapping between the single cladding layers. Different overlap rates can lead to large differences in surface flatness when other process parameters are determined. The greater the flatness, the better the quality

Table 1 Model function of cross-section profile

Function	Function expressions
Parabola function	$y = a * x^2 + b * x + c$
Cosine function	$y = a * \cos(b * x) + c$
Gaussian function	$y = a * e^{-(x-b)^2/c^2}$

Table 2 L125(5³) full factorial experimental design

Factors	Notations	Unit	Level				
			−2	−1	0	1	2
Laser power	P	W	1140	1230	1320	1410	1500
Scanning speed	V	mm/s	10	11	12	13	14
Powder feeding speed	F	r/min	1.9	2	2.1	2.2	2.3

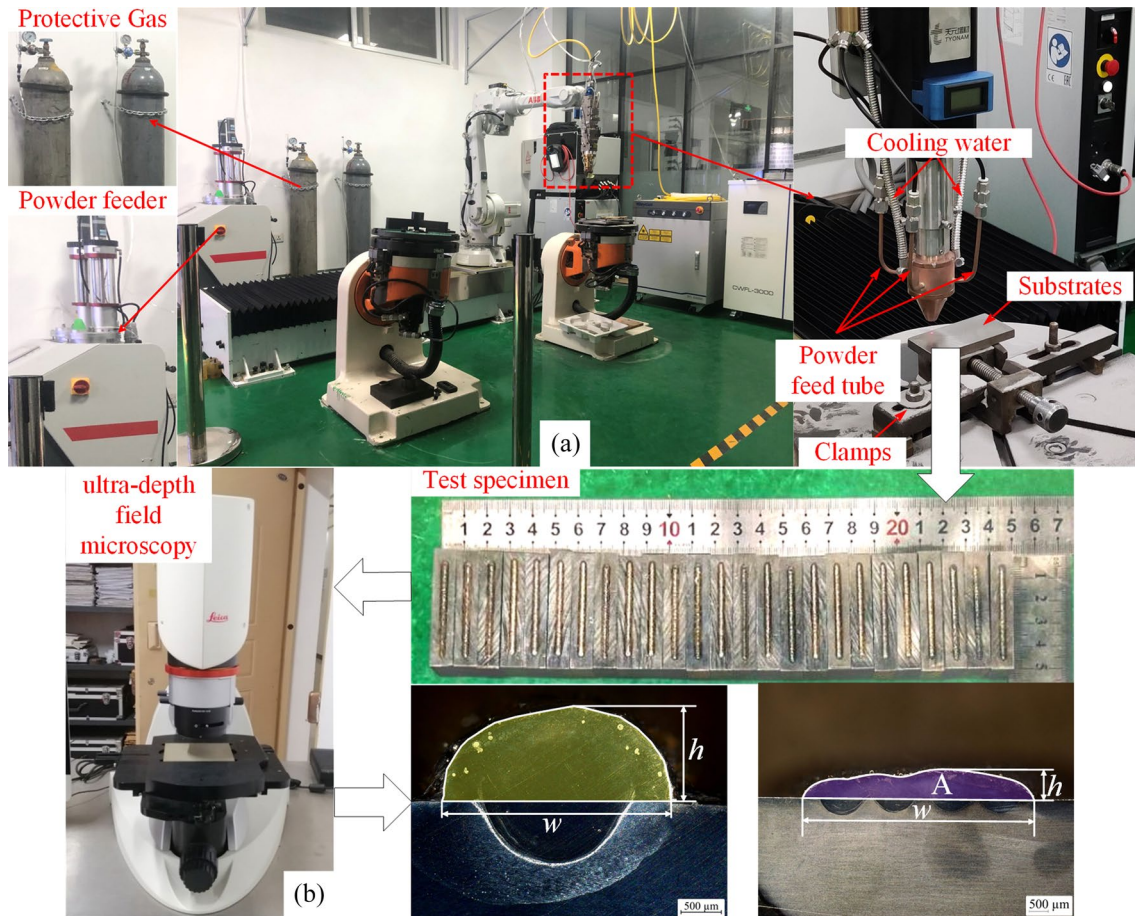


Fig. 2 Laboratory equipment

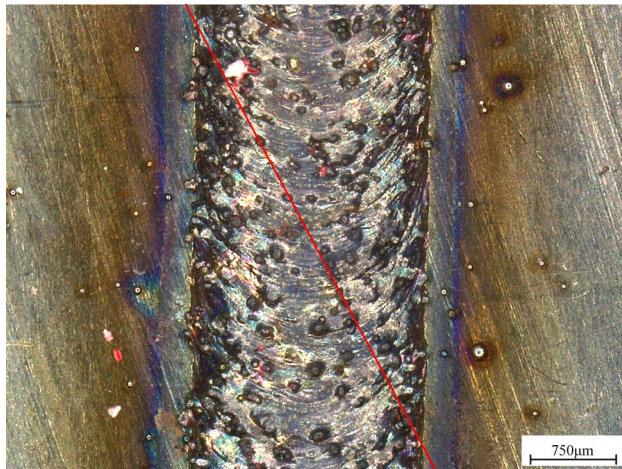


Fig. 3 Surface morphology and the schematic diagram of cross-section profile measurement

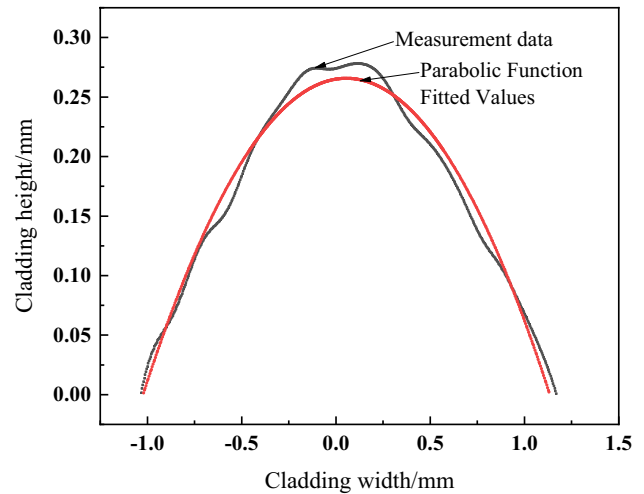


Fig. 4 Curve fitting with a Parabolic function

of the cladding layer and more beneficial it is for subsequent processing. Incorrect selection of the multi-track laser cladding overlap rate can affect the dimensional accuracy

of the part to be repaired or surface modified, and make subsequent processing more difficult. Therefore, controlling the overlap rate λ is a key process parameter in determining

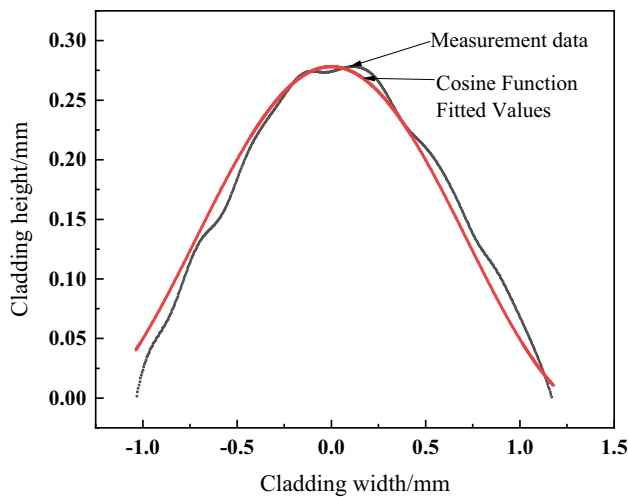


Fig. 5 Curve fitting with a Cosine function

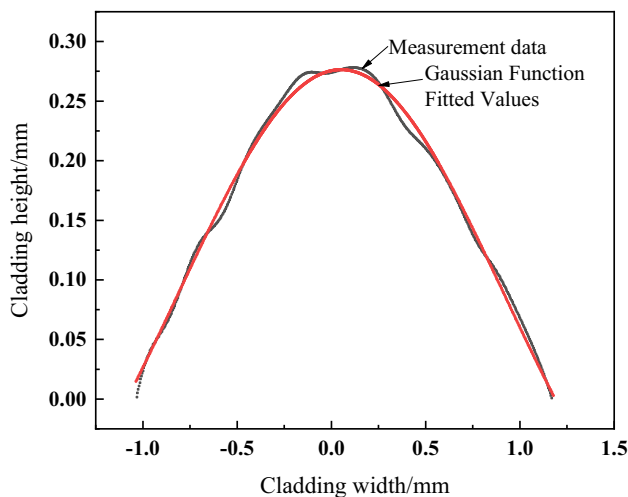


Fig. 6 Curve fitting with Gaussian function

multi-track forming. If the overlap rate $\lambda < 0$, as seen in Fig. 7a, there is no overlap between adjacent laser cladding tracks, and the overlap situation is not overlapped. If the overlap rate is $0 < \lambda < \mu$ (μ is the optimum overlap rate), as seen in Fig. 7b, the overlap situation is an under overlap. The too small remelting area between two single-track cladding layers results in a wave valley with obvious depressions in the overlap region. When the overlap rate is too low, it will increase the surface roughness of the multi-track layer; and will produce holes, cracks and other defects. With the overlap rate λ increasing, molten metal fills the wave valley, the wave valley gradually becomes smaller, and the overlap plane gets closer to the ideal plane. If the overlap rate $\lambda = \mu$, as seen in Fig. 7c, this is the ideal overlap situation. The overlap plane is close to the ideal plane because the molten

metal in the overlap area fills the wave valley exactly and the valley disappears. At this time, the laser cladding layer has a good surface flatness. This overlap situation is beneficial for the laser cladding of the next layer, and it minimizes the error accumulation in the vertical direction. If the overlap ratio $\mu < \lambda < 1$, as shown in Fig. 7d, the overlap situation is excessive. Too much remelting part between two single-track laser cladding layers leads to the wave crest appearing in the overlap region. It will destroy the forming surface of the overlap region and reduce the flatness of the overlap region. At the same time, the increasingly high cladding layer may eventually collide with the laser head, leading to the interruption of the laser cladding process. Therefore, avoiding excessive overlap in the multi-track laser cladding process is necessary.

3.2 Mathematical Model

According to reference [7], the overlap rate equation is chosen as follows.

$$\lambda = 1 - \frac{l}{w} \quad (1)$$

where l is the center distance between two adjacent single-track cladding layers; w represents the cladding width of the single-track cladding layer.

To simplify the mathematical model of multi-track overlap, base on the overlap volume between adjacent single-track cladding layers is equivalent to the filled volume of the multi-track laser cladding overlapping model. The model is usually built based on the following four fundamental assumptions.

- (1) The cross-section profile of the single-track cladding layer is regarded as an ideal Gaussian curve, and the cladding layer maintains a uniform shape and symmetry during the cladding process.
- (2) During multi-track laser cladding, when adjacent single-track cladding layers are overlapped, the excess metal liquid flows freely and fills the wave valleys of adjacent single-track cladding layers.
- (3) The cross-section curvature of the cladding layer remains constant after overlapping.
- (4) The effects of liquid metal surface tension and thermal stress caused by laser rapid cooling are ignored.

The overlapping model is shown in Fig. 8. In the process of multi-track laser cladding, the “excess molten metal” (curve DCE) between the overlapping areas of adjacent single-track cladding layers will spread to the wave valley between adjacent single-track cladding layers (curve ACB). Therefore, when there is a unique distance between two adjacent tracks so that A, G, and B are on a straight line, the overlapping area of

Table 3 Results of three fitting functions

No.	Parabolic functions			Cosine function			Gaussian function		
	Adjusted R^2	SSE	Significance F -value	Adjusted R^2	SSE	Significance F -value	Adjusted R^2	SSE	Significance F -value
1	0.956	0.196	6610.266	0.893	0.483	2502.414	0.957	0.194	35,664.022
2	0.964	0.05	6553.533	0.97	0.041	8015.11	0.982	0.025	269,547.26
3	0.973	0.016	7451.804	0.973	0.016	7462.063	0.976	0.02	108,181.29
4	0.928	0.102	3083.083	0.926	0.105	2987.655	0.892	0.153	14,342.779
5	0.876	0.307	1807.99	0.965	0.086	33,939.229	0.984	0.04	49,615.548
.....									
62	0.93	1.107	4300.975	0.95	0.783	6213.202	0.961	0.608	23,618.284
63	0.971	0.303	10,902.824	0.981	0.193	17,296.233	0.986	0.145	136,907.63
64	0.968	0.082	7562.914	0.962	0.098	6243.766	0.941	0.152	64,663.46
.....									
121	0.983	0.108	16,318.662	0.977	0.148	11,824.493	0.985	0.287	67,188.347
122	0.992	0.02	30,294.069	0.993	0.017	34,871.825	0.995	0.029	280,034.41
123	0.972	0.082	8693.433	0.969	0.091	7806.863	0.951	0.144	72,318.422
124	0.962	0.044	6337.843	0.965	0.04	6856.864	0.966	0.039	157,426.64
125	0.962	0.235	8203.761	0.973	0.17	11,457.83	0.974	0.176	52,719.109

Table 4 Three post-fitting functions

Function	Function expressions
Parabola function	$y = -0.21008x^2 + 1.27108x - 1.54233$
Cosine function	$y = -0.42471 * \cos(1.03862x) - 0.04233$
Gaussian function	$y = 0.76294 * e^{-(x-0.05206)^2/1.51914^2}$

the cladding layer is an ideal plane. Therefore, the area of the curve ACB is equal to the curve DCE.

Since assumption (1), the cross-sectional curve of the cladding layer is uniform and symmetrical, so $S_{ACG} = S_{BCG}$, $S_{DCF} = S_{ECF}$. When AGB is an ideal plane, point C exists such that Eq. (2) holds [8, 9].

$$S_{ACG} = \int_0^{x_c} f(x)dx = S_{ECF} = \int_{x_c}^{w/2} f(x)dx \quad (2)$$

Therefore Eq. (3) holds

$$S_{ACB} = S_{DCE} \quad (3)$$

Point O is set as the original point, and point C is the intersection of the cross-section curves of the two adjacent single-track cladding layers, with the coordinates (x_c, y_c) . The theoretical best overlap rate model is calculated as follows:

Curved-edge triangle $S_{\Delta ABC}$ with Eq. (4)

$$S_{\Delta ABC} = S_{\Delta ACG} + S_{\Delta BCG} = 2S_{\Delta ACG} = 2 \int_0^{x_c} h - a * e^{-(x-b)^2/c^2} dx \quad (4)$$

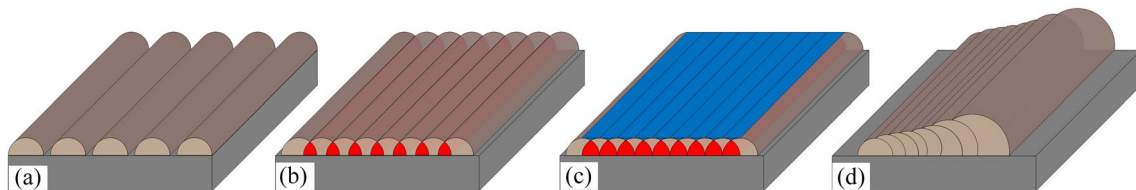
Curved-edge triangle $S_{\Delta ECD}$ with Eq. (5)

$$S_{\Delta ECD} = S_{\Delta DCF} + S_{\Delta ECF} = 2S_{\Delta ECF} = 2 \int_{x_c}^{w/2} a * e^{-(x-b)^2/c^2} dx \quad (5)$$

Based on the Eq. (2)

$$\int_0^{x_c} h - a * e^{-(x-b)^2/c^2} dx = \int_{x_c}^{w/2} a * e^{-(x-b)^2/c^2} dx \quad (6)$$

Solve Eq. (6) to get x_c

**Fig. 7** Results of different overlap rates

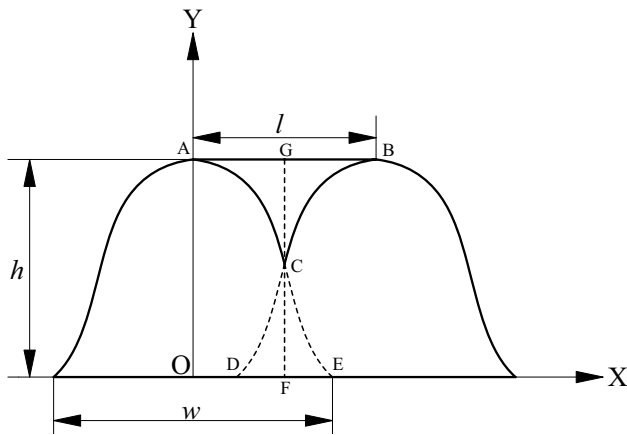


Fig. 8 Principle diagram of the overlapping model

$$x_c = \frac{\sqrt{\pi}ac \left(\operatorname{erf}\left(\frac{b}{c}\right) - \operatorname{erf}\left(\frac{-w+2b}{2c}\right) \right)}{2h} \quad (7)$$

where $\operatorname{erf}(x)$ is the error function, the erf function is defined as Eq. (8)

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (8)$$

According to the above steps, an optimal overlap rate model based on the flat-top overlapping model can be built as Eq. (9):

$$\lambda = 1 - \frac{l}{w} = 1 - \frac{2 * x_c}{w} = 1 - \frac{\sqrt{\pi}ac \left(\operatorname{erf}\left(\frac{b}{c}\right) - \operatorname{erf}\left(\frac{-w+2b}{2c}\right) \right)}{wh} \quad (9)$$

where a , b , c are the coefficients in the function expression in Table 4; w is the cladding width of the single-track cladding layer; h is the cladding height of the single-track cladding layer.

The laser cladding process parameters greatly influence the geometry of the cladding layer. Therefore, the cladding data used in the optimal overlap rate model for the proposed cladding layers should be measured on specimens with optimal single-track cladding process parameters.

4 Experimental Verifications

4.1 Validation Experiments

The 20Cr13 martensitic stainless steel was selected as the substrate for the validation experiments and cut into a specimen of size $100 \times 10 \times 10$ mm for spare. M2 powder, 316L powder and 15-5PH powder were selected as the cladding

Table 5 Morphology data and overlap rate of three materials

Cladding powder	Morphology data and overlap rate		
	Cladding width/mm	Cladding height/mm	Overlap rate (%)
M2	2.693	1.021	39.4
316L	3.057	1.207	51.5
15-5PH	2.993	1.025	42.4

powder. Before the experiment, the surface of the substrate was first polished to remove the surface oxide layer and impurities. And the surface was cleaned with acetone. The ultrasonic cleaning instrument was put in anhydrous ethanol for 30 min, and finally placed in the drying oven at 200°C for drying. After that it was put in the drying cabinet for backup. The powder was put in the drying oven at 200°C for 30 min to ensure dryness and avoid defects such as cracks and porosity in the prepared cladding layer.

The validation experiment parameters are as follows: laser power $P = 1500$ W, scanning speed $V = 11$ mm/s, powder feeding speed $F = 1.9$ r/min. The single-layer multi-track laser cladding verification experiments was conducted to verify the accuracy of the best overlap rate calculation model under the flat-top overlapping model. To ensure the reliability of the data, we conducted three replicate validation experiments with morphological data measured at three locations.

Firstly, single-track laser cladding experiments were conducted for three different powders. The data of their cladding height and width were obtained, and their optimum overlap rates were calculated according to Eq. (7). The data is seen in Table 5.

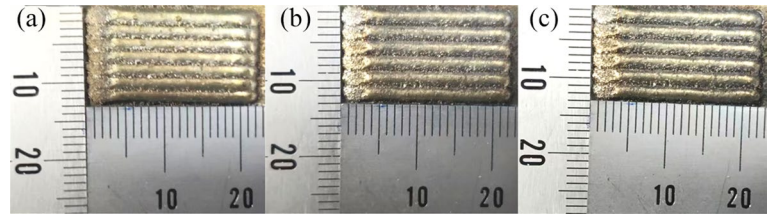
The multi-track cladding layer specimens were prepared, as seen in Fig. 9, with no obvious macroscopic defects, such as fracture, deformation. The prepared specimens were cut in the direction perpendicular to the cladding layer with a metallographic cutter. Afterwards, grinding, polishing and corrosion treatment were carried out. The macroscopic morphology of single-layer multi-track cross-sections was photographed by ultra-deep field microscope.

4.2 Calculation Results of Flatness

To quantitatively describe the forming quality of the multi-track cladding layer, the ratio of the cross-section area of the multi-track cladding layer to the product of the cladding width and cladding height is defined as the surface flatness. Surface flatness definition equation [2], as seen in Eq. (10).

$$\tau = \frac{A}{w \cdot h} \times 100\% \quad (10)$$

Fig. 9 Multi-track cladding layer verification experiment: **a** M2; **b** 316L; **c** 15-5PH



where τ is the surface flatness of the multi-track layer. A is the cross-section area of the multi-track cladding layer. w is the cladding width of the multi-track cladding layer. h is the cladding height of the multi-track cladding layer.

RE is the relative error between the actual surface flatness and the theoretical value. The calculation equation is as follows:

$$RE = \frac{\tau_T - \tau_M}{\tau_M} \times 100\% \quad (11)$$

where τ_M is the actual surface flatness; τ_T is the theoretical value of surface flatness.

The data in the middle area of the cladding layer is collected to reduce the influence of uneven powder feeding on the experimental results. As seen in Fig. 11, the experimental results still have slight wave peaks and wave valleys due to the fact that the flat-top overlapping model is an ideal model, and ignores the effects of liquid surface tension and thermal stress of the metal in the molten state that exist in actual processing. By measuring the data related to the cross-section of the molten cladding layer, the surface flatness of the cladding layer is calculated according to Eq. (8). The measurement schematic is shown in Fig. 10.

The cross-section morphology of the multi-track cladding layers of the three different cladding materials is shown in Fig. 11. There are no obvious wave peaks and wave valleys in the multi-track cladding layers of the three materials. The measured data, the calculated flatness values, and the corresponding relative error for the three different materials of Fig. 11 are shown in Table 6. The surface flatness of the multi-track cladding layer of the three different materials was calculated to be 90.4%, 87.6% and 87.5%. The flatness of the three replicate experiments is shown in Fig. 12. Experimental results show good flatness and minor errors, indicating that the overlapping model benefits the subsequent treatment and processing.

As seen from Table 6, the theoretical value of the flatness of the cladding layer does not reach 100%, which is caused by the single-track cladding layer sectional profile being a Gaussian function. Furthermore, as the overlapping surfaces are considered to be ideally plane, but it is impossible to achieve an ideally plane during the laser cladding process, relative errors are bound to exist.

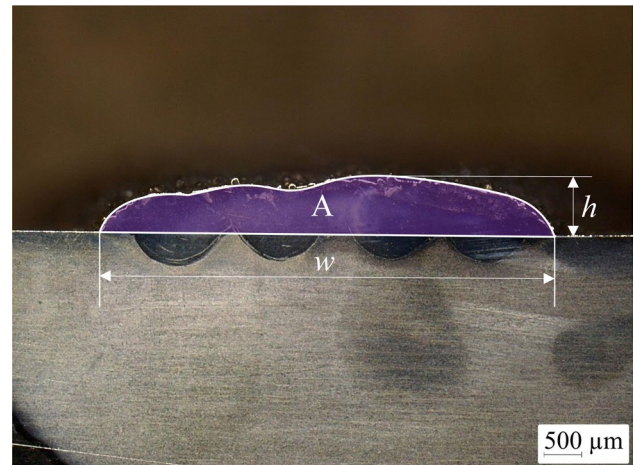


Fig. 10 Schematic diagram of flatness measurement

From the results in Table 6, it can be seen that the flatness of the M2 cladding layer has a large difference compared to 316L and 15-5PH, while the flatness of 316L and 15-5PH is close. The melting points of 20Cr13, M2, 316L and 15-5PH are 1470 °C, 1100 °C, 1400 °C and 1420 °C, respectively. The melting points of 316L and 15-5PH powders are similar, but the melting point of M2 powder is lower than that of 316L and 15-5PH. At the fixed process parameters and the same melting time, the flatness of the M2 cladding layer is better than that of 316L and 15-5PH cladding layers, probably because the surface tension of liquid metal decreases with increasing temperature. So with fixed parameters, the surface tension of M2 with the liquid metal state is lower than 316L with 15-5PH. As the surface tension decreases, the surface energy between the liquid metal and the substrate also decreases, and the wettability increases, resulting in a better wettability of M2 than 316L and 15-5PH. Therefore, the surface flatness of M2 cladding layer is better than 316L and 15-5PH [22–24].

5 Conclusions

A method for optimizing the overlap rate of multi-track laser cladding based on the flat-top model is proposed to improve the flatness of the multi-track cladding layer. The following

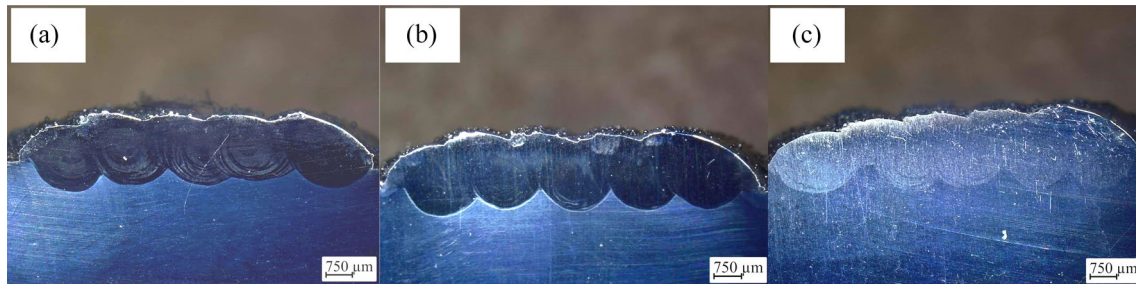


Fig. 11 Cross-section of a multi-track layer of three materials: **a** M2; **b** 316L; **c** 15-5PH

Table 6 Calculation results of multi-track laser cladding

Cladding powder	Cladding area A/mm^2	Cladding width w/mm	Cladding height h/mm	Surface flatness (%)	Theoretical value (%)	Relative error (%)
M2	10.748	9.669	1.230	90.4	96.9	7.1
316L	11.892	9.984	1.360	87.6	97.6	11.4
15-5PH	11.059	10.132	1.248	87.5	97.7	11.6

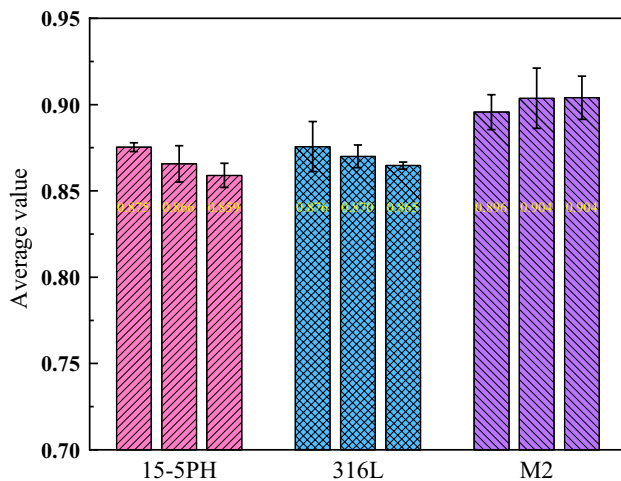


Fig. 12 Comparison of the flatness of the three materials

conclusions can be drawn based on the analysis and experimental study.

- (1) The cross-section profile data were collected by designing a three-factor, five-level full factorial experiment. The Gaussian function was determined to model the multi-track laser cladding overlapping model by comparing the fitting results, as it has a better adjusted R^2 and F -value.
- (2) Based on the overlapping model proposed in the study, the theoretical values of different powder flatness are

96.9%, 97.6% and 97.7%. The results of the validation experiments show that the different cladding powders have good forming morphology. The surface flatness values are 90.4%, 87.6% and 87.5% and the error from the theoretical values are 7.1%, 11.4% and 11.6%, which proves that the model helps to obtain good surface flatness for different cladding powders.

- (3) By verifying the comparison of experimental results. Based on the same overlapping model, the flatness of the cladding layers of different powders is different, but the cladding layers of the powders with similar melting points have similar flatness.

The presented model helps to obtain the optimum overlap rate to improve the surface flatness. However, there are many factors affecting the flatness and there are also coupling effects between the various influencing factors. The next step is to incorporate the optimization model of overlap rate into more process parameters to provide a more accurate model. In addition, more experiments will have to be carried out in the future with various substrates and powders to modify the model.

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